

# Association Analysis Using Apriori, FP-Growth, Sequential Pattern Mining, Infrequent Pattern Mining and Subgraph Pattern Mining

## Introduction

Association analysis is a data mining technique used to discover interesting relationships and patterns among items in large datasets. It is commonly used in market basket analysis to identify products that are frequently purchased together.

In this practical, the Online Retail dataset is used to implement and compare two frequent pattern mining algorithms: Apriori and FP-Growth. Association rules are generated and evaluated using support, confidence, and lift.

The performance of Apriori and FP-Growth is also compared based on the number of frequent itemsets, number of generated rules, and execution time.

## Import Required Libraries

The required Python libraries are imported for data manipulation, visualization, frequent itemset mining, association rule generation, and performance evaluation.

```
In [8]: !pip install mlxtend
```

Requirement already satisfied: mlxtend in C:\Users\legion\AppData\Local\Programs\Python\Python313\Lib\site-packages (0.25.0)

Requirement already satisfied: scipy>=1.16.3 in C:\Users\legion\AppData\Local\Programs\Python\Python313\Lib\site-packages (from mlxtend) (1.16.3)

Requirement already satisfied: numpy>=2.3.5 in C:\Users\legion\AppData\Local\Programs\Python\Python313\Lib\site-packages (from mlxtend) (2.5.2)

Requirement already satisfied: pandas>=2.3.3 in C:\Users\legion\AppData\Local\Programs\Python\Python313\Lib\site-packages (from mlxtend) (3.0.5)

Requirement already satisfied: scikit-learn>=1.8.0 in C:\Users\legion\AppData\Local\Programs\Python\Python313\Lib\site-packages (from mlxtend) (1.8.0)

Requirement already satisfied: matplotlib>=3.10.8 in C:\Users\legion\AppData\Local\Programs\Python\Python313\Lib\site-packages (from mlxtend) (3.10.8)

Requirement already satisfied: joblib>=1.5.2 in C:\Users\legion\AppData\Local\Programs\Python\Python313\Lib\site-packages (from mlxtend) (1.5.3)

Requirement already satisfied: contourpy>=1.0.1 in C:\Users\legion\AppData\Local\Programs\Python\Python313\Lib\site-packages (from matplotlib>=3.10.8->mlxtend) (1.3.3)

Requirement already satisfied: cycler>=0.10 in C:\Users\legion\AppData\Local\Programs\Python\Python313\Lib\site-packages (from matplotlib>=3.10.8->mlxtend) (0.12.1)

Requirement already satisfied: fonttools>=4.22.0 in C:\Users\legion\AppData\Local\Programs\Python\Python313\Lib\site-packages (from matplotlib>=3.10.8->mlxtend) (4.61.1)

Requirement already satisfied: kiwisolver>=1.3.1 in C:\Users\legion\AppData\Local\Programs\Python\Python313\Lib\site-packages (from matplotlib>=3.10.8->mlxtend) (1.4.9)

Requirement already satisfied: packaging>=20.0 in C:\Users\legion\AppData\Roaming\Python\Python313\site-packages (from matplotlib>=3.10.8->mlxtend) (25.0)

Requirement already satisfied: pillow>=8 in C:\Users\legion\AppData\Local\Programs\Python\Python313\Lib\site-packages (from matplotlib>=3.10.8->mlxtend) (11.2.1)

Requirement already satisfied: pyparsing>=3 in C:\Users\legion\AppData\Local\Programs\Python\Python313\Lib\site-packages (from matplotlib>=3.10.8->mlxtend) (3.3.1)

Requirement already satisfied: python-dateutil>=2.7 in C:\Users\legion\AppData\Local\Programs\Python\Python313\Lib\site-packages (from matplotlib>=3.10.8->mlxtend) (2.9.0.post0)

Requirement already satisfied: tzdata in C:\Users\legion\AppData\Local\Programs\Python\Python313\Lib\site-packages (from pandas>=2.3.3->mlxtend) (2025.2)

Requirement already satisfied: six>=1.5 in C:\Users\legion\AppData\Local\Programs\Python\Python313\Lib\site-packages (from python-dateutil>=2.7->matplotlib>=3.10.8->mlxtend) (1.17.0)

Requirement already satisfied: threadpoolctl>=3.2.0 in C:\Users\legion\AppData\Local\Programs\Python\Python313\Lib\site-packages (from scikit-learn>=1.8.0->mlxtend) (3.6.0)

[notice] A new release of pip is available: 26.0.1 -> 26.2.1

[notice] To update, run: python.exe -m pip install --upgrade pip

```
In [9]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import time

from mlxtend.frequent_patterns import apriori, fpgrowth, association_rules

import warnings
warnings.filterwarnings("ignore")
```

## Load the Dataset

```
In [10]: df = pd.read_csv("Online_Retail.csv", encoding="ISO-8859-1")

print("Dataset Shape:", df.shape)
df.head()
```

Dataset Shape: (541909, 8)

```
Out[10]:
```

|   | InvoiceNo | StockCode | Description                                     | Quantity | InvoiceDate     | UnitPrice | CustomerID | Cou         |
|---|-----------|-----------|-------------------------------------------------|----------|-----------------|-----------|------------|-------------|
| 0 | 536365    | 85123A    | WHITE<br>HANGING<br>HEART T-<br>LIGHT<br>HOLDER | 6        | 12/1/10<br>8:26 | 2.55      | 17850.0    | Un<br>Kingc |
| 1 | 536365    | 71053     | WHITE<br>METAL<br>LANTERN                       | 6        | 12/1/10<br>8:26 | 3.39      | 17850.0    | Un<br>Kingc |
| 2 | 536365    | 84406B    | CREAM<br>CUPID<br>HEARTS<br>COAT<br>HANGER      | 8        | 12/1/10<br>8:26 | 2.75      | 17850.0    | Un<br>Kingc |
| 3 | 536365    | 84029G    | KNITTED<br>UNION<br>FLAG HOT<br>WATER<br>BOTTLE | 6        | 12/1/10<br>8:26 | 3.39      | 17850.0    | Un<br>Kingc |
| 4 | 536365    | 84029E    | RED<br>WOOLLY<br>HOTTIE<br>WHITE<br>HEART.      | 6        | 12/1/10<br>8:26 | 3.39      | 17850.0    | Un<br>Kingc |

## Dataset Information

```
In [11]: df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 541909 entries, 0 to 541908
Data columns (total 8 columns):
#   Column          Non-Null Count  Dtype
---  -
0   InvoiceNo        541909 non-null  str
1   StockCode        541909 non-null  str
2   Description      540455 non-null  str
3   Quantity         541909 non-null  int64
4   InvoiceDate       541909 non-null  str
5   UnitPrice        541909 non-null  float64
6   CustomerID       406829 non-null  float64
7   Country          541909 non-null  str
dtypes: float64(2), int64(1), str(5)
memory usage: 66.2 MB
```

```
In [12]: df.describe(include="all")
```

Out[12]:

|        | InvoiceNo | StockCode | Description                                     | Quantity      | InvoiceDate       | UnitPrice     |    |
|--------|-----------|-----------|-------------------------------------------------|---------------|-------------------|---------------|----|
| count  | 541909    | 541909    | 540455                                          | 541909.000000 | 541909            | 541909.000000 | 40 |
| unique | 25900     | 4070      | 4223                                            | NaN           | 23260             | NaN           |    |
| top    | 573585    | 85123A    | WHITE<br>HANGING<br>HEART T-<br>LIGHT<br>HOLDER | NaN           | 10/31/11<br>14:41 | NaN           |    |
| freq   | 1114      | 2313      | 2369                                            | NaN           | 1114              | NaN           |    |
| mean   | NaN       | NaN       | NaN                                             | 9.552250      | NaN               | 4.611114      | 1. |
| std    | NaN       | NaN       | NaN                                             | 218.081158    | NaN               | 96.759853     |    |
| min    | NaN       | NaN       | NaN                                             | -80995.000000 | NaN               | -11062.060000 | 1. |
| 25%    | NaN       | NaN       | NaN                                             | 1.000000      | NaN               | 1.250000      | 1. |
| 50%    | NaN       | NaN       | NaN                                             | 3.000000      | NaN               | 2.080000      | 1. |
| 75%    | NaN       | NaN       | NaN                                             | 10.000000     | NaN               | 4.130000      | 1. |
| max    | NaN       | NaN       | NaN                                             | 80995.000000  | NaN               | 38970.000000  | 1. |

```
In [13]: df.isnull().sum() #missing values analysis
```

```
Out[13]: InvoiceNo      0
        StockCode     0
        Description    1454
        Quantity      0
        InvoiceDate     0
        UnitPrice      0
        CustomerID    135080
        Country        0
        dtype: int64
```

```
In [14]: df.duplicated().sum() #duplicate records
```

```
Out[14]: np.int64(5268)
```

## Transaction Analysis

```
In [15]: df['InvoiceNo'].astype(str).str.startswith('C').sum()
```

```
Out[15]: np.int64(9288)
```

## Data Preprocessing

```
In [16]: # Remove duplicate records
df = df.drop_duplicates()

# Remove cancelled transactions
df = df[~df['InvoiceNo'].astype(str).str.startswith('C')]

# Remove rows with missing product descriptions
df = df.dropna(subset=['Description'])

# Keep only positive quantities and unit prices
df = df[(df['Quantity'] > 0) & (df['UnitPrice'] > 0)]

print("Dataset Shape after preprocessing:", df.shape)
```

Dataset Shape after preprocessing: (524878, 8)

```
In [17]: # Group products by InvoiceNo to create transactions
transactions = df.groupby('InvoiceNo')['Description'].apply(list)

# Check number of transactions
print("Number of transactions:", len(transactions))

# Display first 5 transactions
print("\nFirst 5 transactions:")
print(transactions.head())
```

Number of transactions: 19960

First 5 transactions:

InvoiceNo

|        |                                                   |
|--------|---------------------------------------------------|
| 536365 | [WHITE HANGING HEART T-LIGHT HOLDER, WHITE MET... |
| 536366 | [HAND WARMER UNION JACK, HAND WARMER RED POLKA... |
| 536367 | [ASSORTED COLOUR BIRD ORNAMENT, POPPY'S PLAYHO... |
| 536368 | [JAM MAKING SET WITH JARS, RED COAT RACK PARIS... |
| 536369 | [BATH BUILDING BLOCK WORD]                        |

Name: Description, dtype: object

## Transaction Encoding

Convert transactions into a one-hot encoded format for association rule mining.

```
In [18]: from mlxtend.preprocessing import TransactionEncoder

te = TransactionEncoder()
te_array = te.fit(transactions).transform(transactions)

transaction_df = pd.DataFrame(
    te_array,
    columns=te.columns_
)

print("Transaction matrix shape:", transaction_df.shape)
print(transaction_df.head())
```

Transaction matrix shape: (19960, 4026)

|   | 4 PURPLE FLOCK DINNER CANDLES | 50'S CHRISTMAS GIFT BAG LARGE \ |
|---|-------------------------------|---------------------------------|
| 0 | False                         | False                           |
| 1 | False                         | False                           |
| 2 | False                         | False                           |
| 3 | False                         | False                           |
| 4 | False                         | False                           |

|   | DOLLY GIRL BEAKER | I LOVE LONDON MINI BACKPACK \ |
|---|-------------------|-------------------------------|
| 0 | False             | False                         |
| 1 | False             | False                         |
| 2 | False             | False                         |
| 3 | False             | False                         |
| 4 | False             | False                         |

|   | I LOVE LONDON MINI RUCKSACK | NINE DRAWER OFFICE TIDY \ |
|---|-----------------------------|---------------------------|
| 0 | False                       | False                     |
| 1 | False                       | False                     |
| 2 | False                       | False                     |
| 3 | False                       | False                     |
| 4 | False                       | False                     |

|   | OVAL WALL MIRROR DIAMANTE | RED SPOT GIFT BAG LARGE \ |
|---|---------------------------|---------------------------|
| 0 | False                     | False                     |
| 1 | False                     | False                     |
| 2 | False                     | False                     |
| 3 | False                     | False                     |
| 4 | False                     | False                     |

|   | SET 2 TEA TOWELS I LOVE LONDON | SPACEBOY BABY GIFT SET ... \ |
|---|--------------------------------|------------------------------|
| 0 | False                          | False ...                    |
| 1 | False                          | False ...                    |
| 2 | False                          | False ...                    |
| 3 | False                          | False ...                    |
| 4 | False                          | False ...                    |

|   | ZINC STAR T-LIGHT HOLDER | ZINC SWEETHEART SOAP DISH \ |
|---|--------------------------|-----------------------------|
| 0 | False                    | False                       |
| 1 | False                    | False                       |
| 2 | False                    | False                       |
| 3 | False                    | False                       |
| 4 | False                    | False                       |

|   | ZINC SWEETHEART WIRE LETTER RACK | ZINC T-LIGHT HOLDER STAR LARGE \ |
|---|----------------------------------|----------------------------------|
| 0 | False                            | False                            |
| 1 | False                            | False                            |
| 2 | False                            | False                            |
| 3 | False                            | False                            |
| 4 | False                            | False                            |

|   | ZINC T-LIGHT HOLDER STARS LARGE | ZINC T-LIGHT HOLDER STARS SMALL \ |
|---|---------------------------------|-----------------------------------|
| 0 | False                           | False                             |
| 1 | False                           | False                             |
| 2 | False                           | False                             |
| 3 | False                           | False                             |
| 4 | False                           | False                             |

|   | ZINC TOP 2 DOOR WOODEN SHELF | ZINC WILLIE WINKIE | CANDLE STICK \ |
|---|------------------------------|--------------------|----------------|
| 0 | False                        |                    | False          |
| 1 | False                        |                    | False          |
| 2 | False                        |                    | False          |
| 3 | False                        |                    | False          |
| 4 | False                        |                    | False          |

|   | ZINC WIRE KITCHEN ORGANISER | ZINC WIRE SWEETHEART | LETTER TRAY |
|---|-----------------------------|----------------------|-------------|
| 0 | False                       |                      | False       |
| 1 | False                       |                      | False       |
| 2 | False                       |                      | False       |
| 3 | False                       |                      | False       |
| 4 | False                       |                      | False       |

[5 rows x 4026 columns]

## Frequent Itemset Mining

Apply the Apriori algorithm to identify frequently purchased product combinations.

```
In [19]: from mlxtend.frequent_patterns import apriori

frequent_itemsets = apriori(
    transaction_df,
    min_support=0.01,
    use_colnames=True
)

print("Number of frequent itemsets:", len(frequent_itemsets))
print(frequent_itemsets.head())
```

Number of frequent itemsets: 1902

|   | support  | itemsets                                         |
|---|----------|--------------------------------------------------|
| 0 | 0.013477 | frozenset({ SET 2 TEA TOWELS I LOVE LONDON })    |
| 1 | 0.015932 | frozenset({10 COLOUR SPACEBOY PEN})              |
| 2 | 0.012575 | frozenset({12 MESSAGE CARDS WITH ENVELOPES})     |
| 3 | 0.017786 | frozenset({12 PENCIL SMALL TUBE WOODLAND})       |
| 4 | 0.018136 | frozenset({12 PENCILS SMALL TUBE RED RETROSPOT}) |

## Association Rule Generation

Generate association rules from frequent itemsets using confidence.

```
In [20]: from mlxtend.frequent_patterns import association_rules

rules = association_rules(
    frequent_itemsets,
    metric="confidence",
    min_threshold=0.3
)

print("Number of rules:", len(rules))
print(rules.head())
```

Number of rules: 1824

|   | antecedents \                                | consequents                                     | antecedent support | support  | confidence | lift | representativity | \ |
|---|----------------------------------------------|-------------------------------------------------|--------------------|----------|------------|------|------------------|---|
| 0 | frozenset({SCANDINAVIAN REDS RIBBONS})       | frozenset({6 RIBBONS RUSTIC CHARM})             |                    | 0.022595 |            |      |                  |   |
| 1 | frozenset({60 CAKE CASES DOLLY GIRL DESIGN}) | frozenset({PACK OF 60 PINK PAISLEY CAKE CASES}) |                    | 0.021894 |            |      |                  |   |
| 2 | frozenset({60 CAKE CASES DOLLY GIRL DESIGN}) | frozenset({PACK OF 72 RETROSPOT CAKE CASES})    |                    | 0.021894 |            |      |                  |   |
| 3 | frozenset({60 CAKE CASES VINTAGE CHRISTMAS}) | frozenset({PAPER CHAIN KIT 50'S CHRISTMAS })    |                    | 0.030962 |            |      |                  |   |
| 4 | frozenset({60 CAKE CASES VINTAGE CHRISTMAS}) | frozenset({PAPER CHAIN KIT VINTAGE CHRISTMAS})  |                    | 0.030962 |            |      |                  |   |

|   | consequent support | support  | confidence | lift      | representativity | \ |
|---|--------------------|----------|------------|-----------|------------------|---|
| 0 | 0.047846           | 0.010872 | 0.481153   | 10.056349 | 1.0              |   |
| 1 | 0.043537           | 0.010421 | 0.475973   | 10.932580 | 1.0              |   |
| 2 | 0.066132           | 0.012124 | 0.553776   | 8.373760  | 1.0              |   |
| 3 | 0.058116           | 0.013577 | 0.438511   | 7.545419  | 1.0              |   |
| 4 | 0.041082           | 0.012525 | 0.404531   | 9.846870  | 1.0              |   |

|   | leverage | conviction | zhangs_metric | jaccard  | certainty | kulczynski |
|---|----------|------------|---------------|----------|-----------|------------|
| 0 | 0.009791 | 1.835135   | 0.921379      | 0.182506 | 0.455081  | 0.354189   |
| 1 | 0.009468 | 1.825215   | 0.928867      | 0.189435 | 0.452119  | 0.357664   |
| 2 | 0.010676 | 2.092822   | 0.900290      | 0.159736 | 0.522176  | 0.368555   |
| 3 | 0.011778 | 1.677476   | 0.895186      | 0.179827 | 0.403866  | 0.336066   |
| 4 | 0.011253 | 1.610357   | 0.927151      | 0.210438 | 0.379020  | 0.354704   |

## Rule Evaluation

Filter and rank association rules using confidence and lift.

```
In [21]: # Filter rules with lift greater than 1
strong_rules = rules[rules['lift'] > 1].copy()

# Sort by lift
strong_rules = strong_rules.sort_values(
    by='lift',
    ascending=False
)

print("Number of strong rules:", len(strong_rules))

# Display top 10 rules
strong_rules[
    ['antecedents', 'consequents', 'support', 'confidence', 'lift']
].head(10)
```

Number of strong rules: 1824

Out[21]:

|             | antecedents                                         | consequents                                         | support  | confidence | lift      |
|-------------|-----------------------------------------------------|-----------------------------------------------------|----------|------------|-----------|
| <b>1060</b> | frozenset({HERB MARKER ROSEMARY, HERB MARKER P...}) | frozenset({HERB MARKER THYME})                      | 0.010170 | 0.944186   | 79.518791 |
| <b>1063</b> | frozenset({HERB MARKER THYME})                      | frozenset({HERB MARKER ROSEMARY, HERB MARKER P...}) | 0.010170 | 0.856540   | 79.518791 |
| <b>1061</b> | frozenset({HERB MARKER THYME, HERB MARKER PARS...}) | frozenset({HERB MARKER ROSEMARY})                   | 0.010170 | 0.953052   | 79.262128 |
| <b>1062</b> | frozenset({HERB MARKER ROSEMARY})                   | frozenset({HERB MARKER THYME, HERB MARKER PARS...}) | 0.010170 | 0.845833   | 79.262128 |
| <b>1057</b> | frozenset({HERB MARKER ROSEMARY})                   | frozenset({HERB MARKER BASIL, HERB MARKER THYME})   | 0.010020 | 0.833333   | 79.206349 |
| <b>1054</b> | frozenset({HERB MARKER BASIL, HERB MARKER THYME})   | frozenset({HERB MARKER ROSEMARY})                   | 0.010020 | 0.952381   | 79.206349 |
| <b>1053</b> | frozenset({HERB MARKER BASIL, HERB MARKER ROSE...}) | frozenset({HERB MARKER THYME})                      | 0.010020 | 0.934579   | 78.709728 |
| <b>1058</b> | frozenset({HERB MARKER THYME})                      | frozenset({HERB MARKER BASIL, HERB MARKER ROSE...}) | 0.010020 | 0.843882   | 78.709728 |
| <b>293</b>  | frozenset({HERB MARKER THYME})                      | frozenset({HERB MARKER ROSEMARY})                   | 0.011072 | 0.932489   | 77.552039 |
| <b>292</b>  | frozenset({HERB MARKER ROSEMARY})                   | frozenset({HERB MARKER THYME})                      | 0.011072 | 0.920833   | 77.552039 |

## Top Association Rules

Select strong and interpretable product associations based on support, confidence, and lift.

```
In [22]: # Keep simple one-to-one rules
top_rules = strong_rules[
    (strong_rules['antecedents'].apply(len) == 1) &
    (strong_rules['consequents'].apply(len) == 1)
].copy()

# Sort by lift
top_rules = top_rules.sort_values(
    by='lift',
    ascending=False
)
```

```
print("Number of one-to-one rules:", len(top_rules))

top_rules[
    ['antecedents', 'consequents', 'support', 'confidence', 'lift']
].head(10)
```

Number of one-to-one rules: 779

Out[22]:

|     | antecedents                          | consequents                          | support  | confidence | lift      |
|-----|--------------------------------------|--------------------------------------|----------|------------|-----------|
| 293 | frozenset({HERB MARKER<br>THYME})    | frozenset({HERB MARKER<br>ROSEMARY}) | 0.011072 | 0.932489   | 77.552039 |
| 292 | frozenset({HERB MARKER<br>ROSEMARY}) | frozenset({HERB MARKER<br>THYME})    | 0.011072 | 0.920833   | 77.552039 |
| 290 | frozenset({HERB MARKER<br>THYME})    | frozenset({HERB MARKER<br>PARSLEY})  | 0.010671 | 0.898734   | 75.372833 |
| 291 | frozenset({HERB MARKER<br>PARSLEY})  | frozenset({HERB MARKER<br>THYME})    | 0.010671 | 0.894958   | 75.372833 |
| 288 | frozenset({HERB MARKER<br>ROSEMARY}) | frozenset({HERB MARKER<br>PARSLEY})  | 0.010772 | 0.895833   | 75.129552 |
| 289 | frozenset({HERB MARKER<br>PARSLEY})  | frozenset({HERB MARKER<br>ROSEMARY}) | 0.010772 | 0.903361   | 75.129552 |
| 282 | frozenset({HERB MARKER<br>MINT})     | frozenset({HERB MARKER<br>PARSLEY})  | 0.010621 | 0.883333   | 74.081232 |
| 283 | frozenset({HERB MARKER<br>PARSLEY})  | frozenset({HERB MARKER<br>MINT})     | 0.010621 | 0.890756   | 74.081232 |
| 279 | frozenset({HERB MARKER<br>ROSEMARY}) | frozenset({HERB MARKER<br>BASIL})    | 0.010721 | 0.891667   | 73.544077 |
| 278 | frozenset({HERB MARKER<br>BASIL})    | frozenset({HERB MARKER<br>ROSEMARY}) | 0.010721 | 0.884298   | 73.544077 |

## Final Rule Selection

Select representative association rules while removing reverse-direction duplicates.

```
In [23]: # Remove reverse-direction duplicates
top_rules = top_rules.copy()

top_rules['rule_pair'] = top_rules.apply(
    lambda x: tuple(sorted([
        list(x['antecedents'])[0],
        list(x['consequents'])[0]
    ])),
    axis=1
)

top_rules = top_rules.drop_duplicates('rule_pair')
```

```
# Rank by lift
top_rules = top_rules.sort_values(
    by='lift',
    ascending=False
)

print("Unique one-to-one rules:", len(top_rules))

top_rules[
    ['antecedents', 'consequents', 'support', 'confidence', 'lift']
].head(15)
```

Unique one-to-one rules: 529

Out[23]:

|            | antecedents                                      | consequents                                      | support  | confidence | lift      |
|------------|--------------------------------------------------|--------------------------------------------------|----------|------------|-----------|
| <b>293</b> | frozenset({HERB MARKER THYME})                   | frozenset({HERB MARKER ROSEMARY})                | 0.011072 | 0.932489   | 77.552039 |
| <b>290</b> | frozenset({HERB MARKER THYME})                   | frozenset({HERB MARKER PARSLEY})                 | 0.010671 | 0.898734   | 75.372833 |
| <b>288</b> | frozenset({HERB MARKER ROSEMARY})                | frozenset({HERB MARKER PARSLEY})                 | 0.010772 | 0.895833   | 75.129552 |
| <b>282</b> | frozenset({HERB MARKER MINT})                    | frozenset({HERB MARKER PARSLEY})                 | 0.010621 | 0.883333   | 74.081232 |
| <b>279</b> | frozenset({HERB MARKER ROSEMARY})                | frozenset({HERB MARKER BASIL})                   | 0.010721 | 0.891667   | 73.544077 |
| <b>276</b> | frozenset({HERB MARKER BASIL})                   | frozenset({HERB MARKER PARSLEY})                 | 0.010621 | 0.876033   | 73.468991 |
| <b>284</b> | frozenset({HERB MARKER ROSEMARY})                | frozenset({HERB MARKER MINT})                    | 0.010621 | 0.883333   | 73.463889 |
| <b>280</b> | frozenset({HERB MARKER BASIL})                   | frozenset({HERB MARKER THYME})                   | 0.010521 | 0.867769   | 73.082958 |
| <b>287</b> | frozenset({HERB MARKER MINT})                    | frozenset({HERB MARKER THYME})                   | 0.010321 | 0.858333   | 72.288326 |
| <b>274</b> | frozenset({HERB MARKER BASIL})                   | frozenset({HERB MARKER MINT})                    | 0.010421 | 0.859504   | 71.482094 |
| <b>698</b> | frozenset({REGENCY TEA PLATE PINK})              | frozenset({REGENCY TEA PLATE GREEN })            | 0.013928 | 0.911475   | 47.750785 |
| <b>734</b> | frozenset({SET OF 3 WOODEN STOCKING DECORATION}) | frozenset({SET OF 3 WOODEN TREE DECORATIONS})    | 0.010972 | 0.686520   | 46.767736 |
| <b>692</b> | frozenset({REGENCY MILK JUG PINK })              | frozenset({REGENCY SUGAR BOWL GREEN})            | 0.012174 | 0.738602   | 45.784138 |
| <b>309</b> | frozenset({JAM JAR WITH GREEN LID})              | frozenset({JAM JAR WITH PINK LID})               | 0.011373 | 0.741830   | 43.295111 |
| <b>731</b> | frozenset({SET OF 3 WOODEN SLEIGH DECORATIONS})  | frozenset({SET OF 3 WOODEN STOCKING DECORATION}) | 0.010170 | 0.678930   | 42.480997 |

## Rule Analysis

Examine the strongest association rules using confidence and lift.

```
In [24]: # Top rules by confidence
top_confidence = top_rules.sort_values(
    by='confidence',
    ascending=False
```

```
)

print("Top 10 rules by confidence:")

top_confidence[
    ['antecedents', 'consequents', 'support', 'confidence', 'lift']
].head(10)
```

Top 10 rules by confidence:

Out[24]:

|     | antecedents                                           | consequents                              | support  | confidence | lift      |
|-----|-------------------------------------------------------|------------------------------------------|----------|------------|-----------|
| 52  | frozenset({BEADED<br>CRYSTAL HEART PINK ON<br>STICK}) | frozenset({DOTCOM<br>POSTAGE})           | 0.010070 | 0.975728   | 27.585742 |
| 293 | frozenset({HERB MARKER<br>THYME})                     | frozenset({HERB<br>MARKER ROSEMARY})     | 0.011072 | 0.932489   | 77.552039 |
| 698 | frozenset({REGENCY TEA<br>PLATE PINK})                | frozenset({REGENCY TEA<br>PLATE GREEN }) | 0.013928 | 0.911475   | 47.750785 |
| 290 | frozenset({HERB MARKER<br>THYME})                     | frozenset({HERB<br>MARKER PARSLEY})      | 0.010671 | 0.898734   | 75.372833 |
| 288 | frozenset({HERB MARKER<br>ROSEMARY})                  | frozenset({HERB<br>MARKER PARSLEY})      | 0.010772 | 0.895833   | 75.129552 |
| 279 | frozenset({HERB MARKER<br>ROSEMARY})                  | frozenset({HERB<br>MARKER BASIL})        | 0.010721 | 0.891667   | 73.544077 |
| 284 | frozenset({HERB MARKER<br>ROSEMARY})                  | frozenset({HERB<br>MARKER MINT})         | 0.010621 | 0.883333   | 73.463889 |
| 282 | frozenset({HERB MARKER<br>MINT})                      | frozenset({HERB<br>MARKER PARSLEY})      | 0.010621 | 0.883333   | 74.081232 |
| 703 | frozenset({REGENCY TEA<br>PLATE PINK})                | frozenset({REGENCY TEA<br>PLATE ROSES }) | 0.013477 | 0.881967   | 39.559698 |
| 276 | frozenset({HERB MARKER<br>BASIL})                     | frozenset({HERB<br>MARKER PARSLEY})      | 0.010621 | 0.876033   | 73.468991 |

## Final Association Rules

Filter association rules using support, confidence, and lift to obtain strong and interpretable product associations.

```
In [25]: # Apply final rule thresholds
final_rules = top_rules[
    (top_rules['support'] >= 0.01) &
    (top_rules['confidence'] >= 0.50) &
    (top_rules['lift'] > 1)
].copy()

# Rank by lift
final_rules = final_rules.sort_values(
```

```
        by='lift',  
        ascending=False  
    )  
  
    print("Number of final rules:", len(final_rules))  
  
    final_rules[  
        ['antecedents', 'consequents', 'support', 'confidence', 'lift']  
    ].head(15)
```

Number of final rules: 161

Out [25]:

|            | <b>antecedents</b>                                     | <b>consequents</b>                                     | <b>support</b> | <b>confidence</b> | <b>lift</b> |
|------------|--------------------------------------------------------|--------------------------------------------------------|----------------|-------------------|-------------|
| <b>293</b> | frozenset({HERB MARKER<br>THYME})                      | frozenset({HERB MARKER<br>ROSEMARY})                   | 0.011072       | 0.932489          | 77.552039   |
| <b>290</b> | frozenset({HERB MARKER<br>THYME})                      | frozenset({HERB MARKER<br>PARSLEY})                    | 0.010671       | 0.898734          | 75.372833   |
| <b>288</b> | frozenset({HERB MARKER<br>ROSEMARY})                   | frozenset({HERB MARKER<br>PARSLEY})                    | 0.010772       | 0.895833          | 75.129552   |
| <b>282</b> | frozenset({HERB MARKER<br>MINT})                       | frozenset({HERB MARKER<br>PARSLEY})                    | 0.010621       | 0.883333          | 74.081232   |
| <b>279</b> | frozenset({HERB MARKER<br>ROSEMARY})                   | frozenset({HERB MARKER<br>BASIL})                      | 0.010721       | 0.891667          | 73.544077   |
| <b>276</b> | frozenset({HERB MARKER<br>BASIL})                      | frozenset({HERB MARKER<br>PARSLEY})                    | 0.010621       | 0.876033          | 73.468991   |
| <b>284</b> | frozenset({HERB MARKER<br>ROSEMARY})                   | frozenset({HERB MARKER<br>MINT})                       | 0.010621       | 0.883333          | 73.463889   |
| <b>280</b> | frozenset({HERB MARKER<br>BASIL})                      | frozenset({HERB MARKER<br>THYME})                      | 0.010521       | 0.867769          | 73.082958   |
| <b>287</b> | frozenset({HERB MARKER<br>MINT})                       | frozenset({HERB MARKER<br>THYME})                      | 0.010321       | 0.858333          | 72.288326   |
| <b>274</b> | frozenset({HERB MARKER<br>BASIL})                      | frozenset({HERB MARKER<br>MINT})                       | 0.010421       | 0.859504          | 71.482094   |
| <b>698</b> | frozenset({REGENCY TEA<br>PLATE PINK})                 | frozenset({REGENCY TEA<br>PLATE GREEN })               | 0.013928       | 0.911475          | 47.750785   |
| <b>734</b> | frozenset({SET OF 3<br>WOODEN STOCKING<br>DECORATION}) | frozenset({SET OF 3<br>WOODEN TREE<br>DECORATIONS})    | 0.010972       | 0.686520          | 46.767736   |
| <b>692</b> | frozenset({REGENCY MILK<br>JUG PINK })                 | frozenset({REGENCY<br>SUGAR BOWL GREEN})               | 0.012174       | 0.738602          | 45.784138   |
| <b>309</b> | frozenset({JAM JAR WITH<br>GREEN LID})                 | frozenset({JAM JAR WITH<br>PINK LID})                  | 0.011373       | 0.741830          | 43.295111   |
| <b>731</b> | frozenset({SET OF 3<br>WOODEN SLEIGH<br>DECORATIONS})  | frozenset({SET OF 3<br>WOODEN STOCKING<br>DECORATION}) | 0.010170       | 0.678930          | 42.480997   |

## Association Rule Visualization

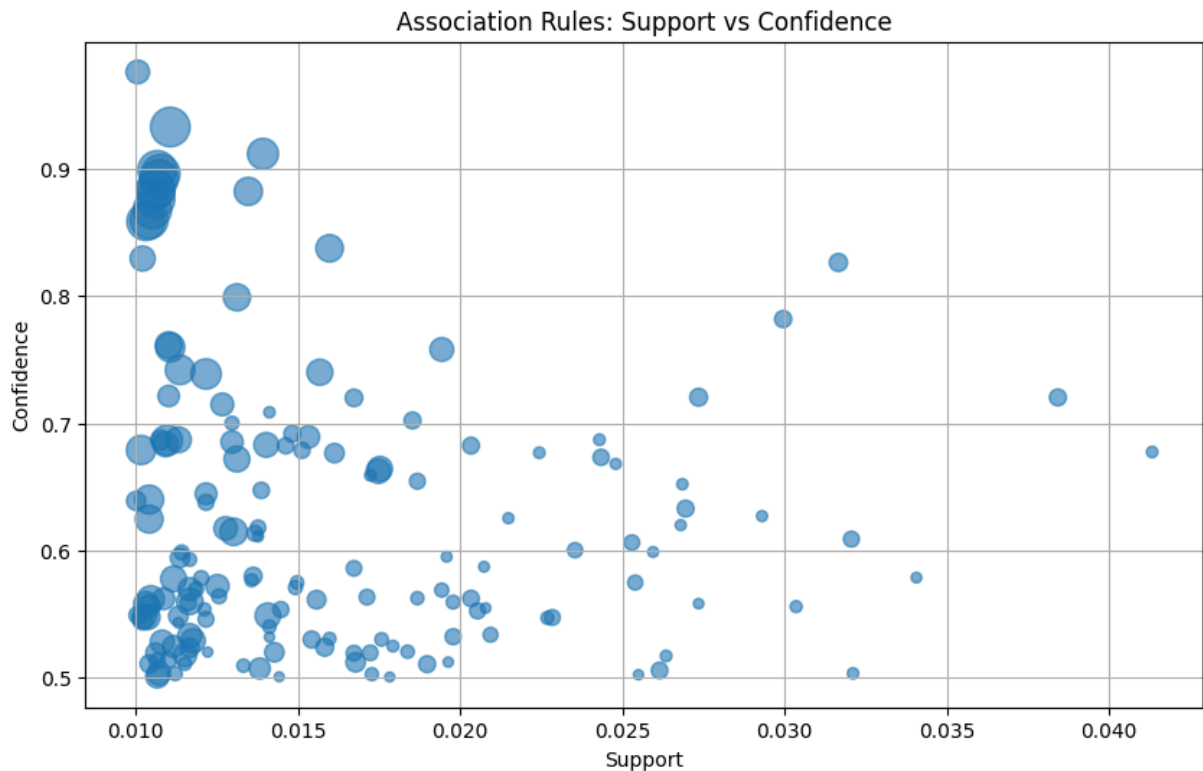
Visualize the relationship between support, confidence, and lift for the final rules.

```
In [26]: import matplotlib.pyplot as plt

plt.figure(figsize=(10, 6))
```

```
plt.scatter(
    final_rules['support'],
    final_rules['confidence'],
    s=final_rules['lift'] * 5,
    alpha=0.6
)

plt.xlabel('Support')
plt.ylabel('Confidence')
plt.title('Association Rules: Support vs Confidence')
plt.grid(True)
plt.show()
```



## FP-Growth

Apply the FP-Growth algorithm to identify frequent itemsets.

```
In [27]: from mlxtend.frequent_patterns import fpgrowth

fp_itemsets = fpgrowth(
    transaction_df,
    min_support=0.01,
    use_colnames=True
)

print("Number of FP-Growth frequent itemsets:", len(fp_itemsets))
print(fp_itemsets.head())
```

```

Number of FP-Growth frequent itemsets: 1902
      support      itemsets
0  0.113026  frozenset({WHITE HANGING HEART T-LIGHT HOLDER})
1  0.022996  frozenset({KNITTED UNION FLAG HOT WATER BOTTLE})
2  0.021493      frozenset({RED WOOLLY HOTTIE WHITE HEART.})
3  0.019038      frozenset({SET 7 BABUSHKA NESTING BOXES})
4  0.015381      frozenset({WHITE METAL LANTERN})

```

## FP-Tree Construction

FP-Growth uses an FP-Tree (Frequent Pattern Tree) to represent the transaction database in a compact tree structure.

Unlike Apriori, which repeatedly generates and tests candidate itemsets, FP-Growth compresses the transactions into an FP-Tree and mines frequent patterns directly from this structure.

The main steps in FP-Tree construction are:

1. Count the frequency of each item.
2. Remove items that do not satisfy the minimum support threshold.
3. Sort items in each transaction according to their frequency.
4. Insert the ordered transactions into the FP-Tree.
5. Share common prefixes between transactions.
6. Use the resulting tree to mine frequent itemsets.

The following implementation demonstrates the basic construction of an FP-Tree using the same transaction data used for Apriori and FP-Growth.

```

In [28]: from collections import Counter

# Use the transaction lists created earlier
# transactions should contain lists of products for each invoice

min_support_count = int(0.01 * len(transactions))

# Count item frequencies
item_counts = Counter(
    item
    for transaction in transactions
    for item in transaction
)

# Keep only frequent items
frequent_items = {
    item: count
    for item, count in item_counts.items()
    if count >= min_support_count
}

# Sort frequent items by frequency

```

```
sorted_frequent_items = sorted(
    frequent_items.items(),
    key=lambda x: x[1],
    reverse=True
)

print("Minimum support count:", min_support_count)
print("Number of frequent items:", len(frequent_items))

print("\nTop 10 frequent items:")
for item, count in sorted_frequent_items[:10]:
    print(f"{item}: {count}")
```

Minimum support count: 199

Number of frequent items: 830

Top 10 frequent items:

WHITE HANGING HEART T-LIGHT HOLDER: 2311

JUMBO BAG RED RETROSPOT: 2109

REGENCY CAKESTAND 3 TIER: 2007

PARTY BUNTING: 1699

LUNCH BAG RED RETROSPOT: 1581

ASSORTED COLOUR BIRD ORNAMENT: 1476

SET OF 3 CAKE TINS PANTRY DESIGN : 1392

PACK OF 72 RETROSPOT CAKE CASES: 1352

LUNCH BAG BLACK SKULL.: 1301

NATURAL SLATE HEART CHALKBOARD : 1255

## FP-Tree Node Structure

Each node of an FP-Tree stores:

- The item name
- The frequency/count of the item
- A reference to its parent
- Its child nodes

A common prefix shared by multiple transactions is represented only once, which provides the main memory advantage of the FP-Tree representation.

```
In [29]: class FPNode:
    def __init__(self, item=None, count=0, parent=None):
        self.item = item
        self.count = count
        self.parent = parent
        self.children = {}

    def add_child(self, item):
        if item not in self.children:
            self.children[item] = FPNode(
                item=item,
                count=0,
                parent=self
            )
```

```

        return self.children[item]

# Create root node
fp_root = FPNode("ROOT")

# Insert transactions into FP-Tree
for transaction in transactions:

    # Keep only frequent items
    filtered_transaction = [
        item for item in transaction
        if item in frequent_items
    ]

    # Sort according to global frequency
    filtered_transaction.sort(
        key=lambda item: frequent_items[item],
        reverse=True
    )

    # Insert into tree
    current_node = fp_root

    for item in filtered_transaction:
        current_node = current_node.add_child(item)
        current_node.count += 1

print("FP-Tree constructed successfully.")
print("Number of root-level branches:", len(fp_root.children))

```

FP-Tree constructed successfully.  
 Number of root-level branches: 647

```

In [51]: def print_tree(node, level=0, max_depth=3, max_children=5):
        if level >= max_depth:
            return

        # Display only the first few children at each level
        children = list(node.children.values())[:max_children]

        for child in children:
            print(
                " " * level +
                f"└─ {child.item} ({child.count})"
            )

            print_tree(
                child,
                level + 1,
                max_depth,
                max_children
            )

```

```
print("Sample FP-Tree Structure")
print("=" * 30)

print_tree(
    fp_root,
    max_depth=3,
    max_children=5
)
```

## Sample FP-Tree Structure

```

=====
└─ WHITE HANGING HEART T-LIGHT HOLDER (2256)
    └─ KNITTED UNION FLAG HOT WATER BOTTLE (1)
        └─ RED WOOLLY HOTTIE WHITE HEART. (1)
    └─ WOODEN PICTURE FRAME WHITE FINISH (65)
        └─ WOODEN FRAME ANTIQUE WHITE (39)
        └─ HOME BUILDING BLOCK WORD (2)
        └─ RED RETROSPOT CHARLOTTE BAG (2)
        └─ WOOD 2 DRAWER CABINET WHITE FINISH (3)
        └─ SET OF 4 PANTRY JELLY MOULDS (2)
    └─ JUMBO BAG RED RETROSPOT (436)
        └─ JAM MAKING SET PRINTED (2)
        └─ LUNCH BAG BLACK SKULL. (13)
        └─ REGENCY CAKESTAND 3 TIER (104)
        └─ STRAWBERRY CERAMIC TRINKET BOX (1)
        └─ PACK OF 72 RETROSPOT CAKE CASES (18)
    └─ CLOTHES PEGS RETROSPOT PACK 24 (6)
        └─ HAND OVER THE CHOCOLATE SIGN (1)
        └─ MEMO BOARD RETROSPOT DESIGN (1)
        └─ GIN + TONIC DIET METAL SIGN (1)
        └─ RED HANGING HEART T-LIGHT HOLDER (1)
        └─ WOODEN HEART CHRISTMAS SCANDINAVIAN (1)
    └─ LUNCH BAG RED RETROSPOT (110)
        └─ JAM MAKING SET PRINTED (3)
        └─ LUNCH BAG BLACK SKULL. (31)
        └─ LUNCH BAG RED RETROSPOT (1)
        └─ ASSORTED COLOUR BIRD ORNAMENT (9)
        └─ PAPER CHAIN KIT 50'S CHRISTMAS (2)
    └─ HAND WARMER UNION JACK (20)
        └─ HAND WARMER RED RETROSPOT (1)
        └─ PLAYING CARDS KEEP CALM & CARRY ON (1)
    └─ ASSORTED COLOUR BIRD ORNAMENT (700)
        └─ HOME BUILDING BLOCK WORD (14)
            └─ LOVE BUILDING BLOCK WORD (2)
            └─ SCOTTIE DOG HOT WATER BOTTLE (1)
            └─ SMALL GLASS HEART TRINKET POT (1)
            └─ DOORMAT RED RETROSPOT (1)
            └─ WOOD BLACK BOARD ANT WHITE FINISH (2)
        └─ WOODEN BOX OF DOMINOES (1)
            └─ 3 STRIPEY MICE FELTCRAFT (1)
        └─ PACK OF 72 RETROSPOT CAKE CASES (34)
            └─ ALARM CLOCK BAKELIKE RED (1)
            └─ VICTORIAN GLASS HANGING T-LIGHT (2)
            └─ PAPER CHAIN KIT 50'S CHRISTMAS (2)
            └─ HEART OF WICKER LARGE (1)
            └─ ROSES REGENCY TEACUP AND SAUCER (2)
        └─ PACK OF 60 PINK PAISLEY CAKE CASES (5)
            └─ RETROSPOT HEART HOT WATER BOTTLE (2)
            └─ GIN + TONIC DIET METAL SIGN (1)
            └─ PAPER CHAIN KIT VINTAGE CHRISTMAS (2)
        └─ RETROSPOT HEART HOT WATER BOTTLE (2)
            └─ WOODEN HEART CHRISTMAS SCANDINAVIAN (1)
    └─ JAM MAKING SET WITH JARS (150)
        └─ DOORMAT RED RETROSPOT (1)
        └─ CHILLI LIGHTS (1)

```

```

└─ ALARM CLOCK BAKELIKE RED (2)
    └─ ALARM CLOCK BAKELIKE GREEN (1)
        └─ DOLLY GIRL LUNCH BOX (1)
└─ HAND WARMER OWL DESIGN (2)
    └─ WOODEN BOX OF DOMINOES (1)
        └─ HAND WARMER BIRD DESIGN (1)
└─ REX CASH+CARRY JUMBO SHOPPER (2)
    └─ VINTAGE HEADS AND TAILS CARD GAME (1)
        └─ CHILLI LIGHTS (1)
└─ GIN + TONIC DIET METAL SIGN (2)
    └─ CHARLOTTE BAG PINK POLKADOT (1)
        └─ IVORY KITCHEN SCALES (1)
└─ BATH BUILDING BLOCK WORD (3)

```

## FP-Tree Observation

The FP-Tree stores transactions using shared prefixes. Therefore, transactions containing common high-frequency items can share the same tree branches.

This compact representation allows FP-Growth to mine frequent patterns without the repeated candidate generation used by Apriori.

## Comparison of Apriori and FP-Growth Frequent Itemsets

Both Apriori and FP-Growth were applied to the same transaction matrix using a minimum support threshold of 0.01. The frequent itemsets generated by both algorithms are compared to verify whether they identify the same patterns.

```

In [31]: # Convert itemsets from both algorithms into sets for comparison

apriori_itemsets_set = set(
    frozenset(itemset) for itemset in frequent_itemsets['itemsets']
)

fpgrowth_itemsets_set = set(
    frozenset(itemset) for itemset in fp_itemsets['itemsets']
)

# Compare the itemsets
common_itemsets = apriori_itemsets_set & fpgrowth_itemsets_set
only_apriori = apriori_itemsets_set - fpgrowth_itemsets_set
only_fpgrowth = fpgrowth_itemsets_set - apriori_itemsets_set

print("Apriori frequent itemsets:", len(apriori_itemsets_set))
print("FP-Growth frequent itemsets:", len(fpgrowth_itemsets_set))
print("Common frequent itemsets:", len(common_itemsets))
print("Only in Apriori:", len(only_apriori))
print("Only in FP-Growth:", len(only_fpgrowth))

```

Apriori frequent itemsets: 1902  
 FP-Growth frequent itemsets: 1902  
 Common frequent itemsets: 1902  
 Only in Apriori: 0  
 Only in FP-Growth: 0

## Association Rule Generation using FP-Growth

Association rules are generated from the frequent itemsets obtained using FP-Growth. The minimum confidence threshold is kept consistent with the Apriori analysis. The resulting rules will later be filtered and compared with the Apriori rules.

```
In [32]: from mlxtend.frequent_patterns import association_rules

fp_rules = association_rules(
    fp_itemsets,
    metric="confidence",
    min_threshold=0.0
)

print("Number of FP-Growth association rules:", len(fp_rules))

print(fp_rules[
    ['antecedents', 'consequents', 'support', 'confidence', 'lift']
].head())
```

```
Number of FP-Growth association rules: 3360
                                antecedents \
0                                frozenset({HAND WARMER OWL DESIGN})
1                                frozenset({HAND WARMER UNION JACK})
2                                frozenset({HAND WARMER SCOTTY DOG DESIGN})
3                                frozenset({HAND WARMER UNION JACK})
4                                frozenset({WHITE HANGING HEART T-LIGHT HOLDER})

                                consequents  support  confidence  lift
0                                frozenset({HAND WARMER UNION JACK})  0.011824    0.360856  15.036933
1                                frozenset({HAND WARMER OWL DESIGN})  0.011824    0.492693  15.036933
2                                frozenset({HAND WARMER UNION JACK})  0.010772    0.392336  16.348689
3                                frozenset({HAND WARMER SCOTTY DOG DESIGN})  0.010772    0.448852  16.348689
4                                frozenset({ASSORTED COLOUR BIRD ORNAMENT})  0.016182    0.143174   1.964088
```

## Filtering FP-Growth Association Rules

The FP-Growth association rules are filtered using the same confidence threshold and rule-selection criteria applied during the Apriori analysis. This ensures that the comparison between Apriori and FP-Growth is fair and consistent.

```
In [33]: # Generate FP-Growth association rules
          # using the same confidence threshold as Apriori

fp_rules = association_rules(
    fp_itemsets,
```

```

    metric="confidence",
    min_threshold=0.3
)

print("Number of FP-Growth rules:", len(fp_rules))

print(
    fp_rules[
        ['antecedents', 'consequents', 'support', 'confidence', 'lift']
    ].head()
)

```

Number of FP-Growth rules: 1824

|   | antecedents \                              |  | consequents | support | confidence \ |
|---|--------------------------------------------|--|-------------|---------|--------------|
| 0 | frozenset({HAND WARMER OWL DESIGN})        |  |             |         |              |
| 1 | frozenset({HAND WARMER UNION JACK})        |  |             |         |              |
| 2 | frozenset({HAND WARMER SCOTTY DOG DESIGN}) |  |             |         |              |
| 3 | frozenset({HAND WARMER UNION JACK})        |  |             |         |              |
| 4 | frozenset({HOME BUILDING BLOCK WORD})      |  |             |         |              |

|   |                                                 | consequents | support  | confidence \ |
|---|-------------------------------------------------|-------------|----------|--------------|
| 0 | frozenset({HAND WARMER UNION JACK})             |             | 0.011824 | 0.360856     |
| 1 | frozenset({HAND WARMER OWL DESIGN})             |             | 0.011824 | 0.492693     |
| 2 | frozenset({HAND WARMER UNION JACK})             |             | 0.010772 | 0.392336     |
| 3 | frozenset({HAND WARMER SCOTTY DOG DESIGN})      |             | 0.010772 | 0.448852     |
| 4 | frozenset({WHITE HANGING HEART T-LIGHT HOLDER}) |             | 0.012776 | 0.328185     |

|   | lift      |
|---|-----------|
| 0 | 15.036933 |
| 1 | 15.036933 |
| 2 | 16.348689 |
| 3 | 16.348689 |
| 4 | 2.903626  |

## Strong FP-Growth Association Rules

Rules with a lift greater than 1 are considered strong positive associations. The same criterion used in the Apriori analysis is applied to the FP-Growth rules.

```

In [34]: # Filter FP-Growth rules with lift greater than 1
fp_strong_rules = fp_rules[fp_rules['lift'] > 1].copy()

# Sort by lift
fp_strong_rules = fp_strong_rules.sort_values(
    by='lift',
    ascending=False
)

print("Number of strong FP-Growth rules:", len(fp_strong_rules))

fp_strong_rules[
    ['antecedents', 'consequents', 'support', 'confidence', 'lift']
].head(10)

```

Number of strong FP-Growth rules: 1824

Out [34]:

|            | antecedents                                             | consequents                                             | support  | confidence | lift      |
|------------|---------------------------------------------------------|---------------------------------------------------------|----------|------------|-----------|
| <b>818</b> | frozenset({HERB MARKER<br>ROSEMARY, HERB<br>MARKER P... | frozenset({HERB MARKER<br>THYME})                       | 0.010170 | 0.944186   | 79.518791 |
| <b>821</b> | frozenset({HERB MARKER<br>THYME})                       | frozenset({HERB MARKER<br>ROSEMARY, HERB<br>MARKER P... | 0.010170 | 0.856540   | 79.518791 |
| <b>820</b> | frozenset({HERB MARKER<br>ROSEMARY})                    | frozenset({HERB MARKER<br>THYME, HERB MARKER<br>PARS... | 0.010170 | 0.845833   | 79.262128 |
| <b>819</b> | frozenset({HERB MARKER<br>THYME, HERB MARKER<br>PARS... | frozenset({HERB MARKER<br>ROSEMARY})                    | 0.010170 | 0.953052   | 79.262128 |
| <b>827</b> | frozenset({HERB MARKER<br>ROSEMARY})                    | frozenset({HERB MARKER<br>BASIL, HERB MARKER<br>THYME}) | 0.010020 | 0.833333   | 79.206349 |
| <b>824</b> | frozenset({HERB MARKER<br>BASIL, HERB MARKER<br>THYME}) | frozenset({HERB MARKER<br>ROSEMARY})                    | 0.010020 | 0.952381   | 79.206349 |
| <b>823</b> | frozenset({HERB MARKER<br>BASIL, HERB MARKER<br>ROSE... | frozenset({HERB MARKER<br>THYME})                       | 0.010020 | 0.934579   | 78.709728 |
| <b>828</b> | frozenset({HERB MARKER<br>THYME})                       | frozenset({HERB MARKER<br>BASIL, HERB MARKER<br>ROSE... | 0.010020 | 0.843882   | 78.709728 |
| <b>809</b> | frozenset({HERB MARKER<br>ROSEMARY})                    | frozenset({HERB MARKER<br>THYME})                       | 0.011072 | 0.920833   | 77.552039 |
| <b>810</b> | frozenset({HERB MARKER<br>THYME})                       | frozenset({HERB MARKER<br>ROSEMARY})                    | 0.011072 | 0.932489   | 77.552039 |

## One-to-One FP-Growth Association Rules

To focus on direct item-to-item associations, rules are filtered so that both the antecedent and consequent contain exactly one item. This is the same filtering criterion used in the Apriori analysis.

In [35]:

```
# Keep only one-to-one rules
fp_one_to_one_rules = fp_strong_rules[
    (fp_strong_rules['antecedents'].apply(len) == 1) &
    (fp_strong_rules['consequents'].apply(len) == 1)
].copy()

print("Number of one-to-one FP-Growth rules:", len(fp_one_to_one_rules))
```

```
fp_one_to_one_rules[
    ['antecedents', 'consequents', 'support', 'confidence', 'lift']
].head(10)
```

Number of one-to-one FP-Growth rules: 779

Out[35]:

|            | antecedents                       | consequents                       | support  | confidence | lift      |
|------------|-----------------------------------|-----------------------------------|----------|------------|-----------|
| <b>809</b> | frozenset({HERB MARKER ROSEMARY}) | frozenset({HERB MARKER THYME})    | 0.011072 | 0.920833   | 77.552039 |
| <b>810</b> | frozenset({HERB MARKER THYME})    | frozenset({HERB MARKER ROSEMARY}) | 0.011072 | 0.932489   | 77.552039 |
| <b>811</b> | frozenset({HERB MARKER THYME})    | frozenset({HERB MARKER PARSLEY})  | 0.010671 | 0.898734   | 75.372833 |
| <b>812</b> | frozenset({HERB MARKER PARSLEY})  | frozenset({HERB MARKER THYME})    | 0.010671 | 0.894958   | 75.372833 |
| <b>804</b> | frozenset({HERB MARKER PARSLEY})  | frozenset({HERB MARKER ROSEMARY}) | 0.010772 | 0.903361   | 75.129552 |
| <b>803</b> | frozenset({HERB MARKER ROSEMARY}) | frozenset({HERB MARKER PARSLEY})  | 0.010772 | 0.895833   | 75.129552 |
| <b>805</b> | frozenset({HERB MARKER MINT})     | frozenset({HERB MARKER PARSLEY})  | 0.010621 | 0.883333   | 74.081232 |
| <b>806</b> | frozenset({HERB MARKER PARSLEY})  | frozenset({HERB MARKER MINT})     | 0.010621 | 0.890756   | 74.081232 |
| <b>798</b> | frozenset({HERB MARKER ROSEMARY}) | frozenset({HERB MARKER BASIL})    | 0.010721 | 0.891667   | 73.544077 |
| <b>797</b> | frozenset({HERB MARKER BASIL})    | frozenset({HERB MARKER ROSEMARY}) | 0.010721 | 0.884298   | 73.544077 |

## Removing Duplicate One-to-One FP-Growth Rules

The one-to-one rules contain reciprocal relationships, where the same pair of products appears in both directions. Duplicate item pairs are removed while retaining a single representative rule for each unique pair, following the same procedure used for Apriori.

```
In [36]: # Create a unique pair identifier for each rule
fp_one_to_one_rules['rule_pair'] = fp_one_to_one_rules.apply(
    lambda row: frozenset(
        [next(iter(row['antecedents'])),
         next(iter(row['consequents']))]),
    ),
    axis=1
)

# Keep one rule for each unique item pair
fp_unique_rules = fp_one_to_one_rules.drop_duplicates(
    subset='rule_pair'
```

```
).copy()

# Sort by lift
fp_unique_rules = fp_unique_rules.sort_values(
    by='lift',
    ascending=False
)

print("Unique one-to-one FP-Growth rules:", len(fp_unique_rules))

fp_unique_rules[
    ['antecedents', 'consequents', 'support', 'confidence', 'lift']
].head(15)
```

Unique one-to-one FP-Growth rules: 529

Out[36]:

|             | antecedents                                      | consequents                                      | support  | confidence | lift      |
|-------------|--------------------------------------------------|--------------------------------------------------|----------|------------|-----------|
| <b>809</b>  | frozenset({HERB MARKER ROSEMARY})                | frozenset({HERB MARKER THYME})                   | 0.011072 | 0.920833   | 77.552039 |
| <b>811</b>  | frozenset({HERB MARKER THYME})                   | frozenset({HERB MARKER PARSLEY})                 | 0.010671 | 0.898734   | 75.372833 |
| <b>804</b>  | frozenset({HERB MARKER PARSLEY})                 | frozenset({HERB MARKER ROSEMARY})                | 0.010772 | 0.903361   | 75.129552 |
| <b>805</b>  | frozenset({HERB MARKER MINT})                    | frozenset({HERB MARKER PARSLEY})                 | 0.010621 | 0.883333   | 74.081232 |
| <b>798</b>  | frozenset({HERB MARKER ROSEMARY})                | frozenset({HERB MARKER BASIL})                   | 0.010721 | 0.891667   | 73.544077 |
| <b>807</b>  | frozenset({HERB MARKER BASIL})                   | frozenset({HERB MARKER PARSLEY})                 | 0.010621 | 0.876033   | 73.468991 |
| <b>800</b>  | frozenset({HERB MARKER MINT})                    | frozenset({HERB MARKER ROSEMARY})                | 0.010621 | 0.883333   | 73.463889 |
| <b>813</b>  | frozenset({HERB MARKER BASIL})                   | frozenset({HERB MARKER THYME})                   | 0.010521 | 0.867769   | 73.082958 |
| <b>815</b>  | frozenset({HERB MARKER THYME})                   | frozenset({HERB MARKER MINT})                    | 0.010321 | 0.869198   | 72.288326 |
| <b>801</b>  | frozenset({HERB MARKER BASIL})                   | frozenset({HERB MARKER MINT})                    | 0.010421 | 0.859504   | 71.482094 |
| <b>1756</b> | frozenset({REGENCY TEA PLATE GREEN })            | frozenset({REGENCY TEA PLATE PINK})              | 0.013928 | 0.729659   | 47.750785 |
| <b>1782</b> | frozenset({SET OF 3 WOODEN STOCKING DECORATION}) | frozenset({SET OF 3 WOODEN TREE DECORATIONS})    | 0.010972 | 0.686520   | 46.767736 |
| <b>1751</b> | frozenset({REGENCY MILK JUG PINK })              | frozenset({REGENCY SUGAR BOWL GREEN})            | 0.012174 | 0.738602   | 45.784138 |
| <b>451</b>  | frozenset({JAM JAR WITH PINK LID})               | frozenset({JAM JAR WITH GREEN LID})              | 0.011373 | 0.663743   | 43.295111 |
| <b>1779</b> | frozenset({SET OF 3 WOODEN SLEIGH DECORATIONS})  | frozenset({SET OF 3 WOODEN STOCKING DECORATION}) | 0.010170 | 0.678930   | 42.480997 |

## Final FP-Growth Rule Selection

The final FP-Growth rules are selected using the same thresholds applied to the Apriori rules: minimum support of 0.01, minimum confidence of 0.50, and lift greater than 1. This ensures that both algorithms are evaluated using identical rule-selection criteria.

```
In [37]: # Apply the same final rule thresholds used for Apriori
fp_final_rules = fp_unique_rules[
    (fp_unique_rules['support'] >= 0.01) &
    (fp_unique_rules['confidence'] >= 0.50) &
    (fp_unique_rules['lift'] > 1)
].copy()

# Rank by lift
fp_final_rules = fp_final_rules.sort_values(
    by='lift',
    ascending=False
)

print("Number of final FP-Growth rules:", len(fp_final_rules))

fp_final_rules[
    ['antecedents', 'consequents', 'support', 'confidence', 'lift']
].head(15)
```

Number of final FP-Growth rules: 157

Out[37]:

|             | antecedents                                      | consequents                                      | support  | confidence | lift      |
|-------------|--------------------------------------------------|--------------------------------------------------|----------|------------|-----------|
| <b>809</b>  | frozenset({HERB MARKER ROSEMARY})                | frozenset({HERB MARKER THYME})                   | 0.011072 | 0.920833   | 77.552039 |
| <b>811</b>  | frozenset({HERB MARKER THYME})                   | frozenset({HERB MARKER PARSLEY})                 | 0.010671 | 0.898734   | 75.372833 |
| <b>804</b>  | frozenset({HERB MARKER PARSLEY})                 | frozenset({HERB MARKER ROSEMARY})                | 0.010772 | 0.903361   | 75.129552 |
| <b>805</b>  | frozenset({HERB MARKER MINT})                    | frozenset({HERB MARKER PARSLEY})                 | 0.010621 | 0.883333   | 74.081232 |
| <b>798</b>  | frozenset({HERB MARKER ROSEMARY})                | frozenset({HERB MARKER BASIL})                   | 0.010721 | 0.891667   | 73.544077 |
| <b>807</b>  | frozenset({HERB MARKER BASIL})                   | frozenset({HERB MARKER PARSLEY})                 | 0.010621 | 0.876033   | 73.468991 |
| <b>800</b>  | frozenset({HERB MARKER MINT})                    | frozenset({HERB MARKER ROSEMARY})                | 0.010621 | 0.883333   | 73.463889 |
| <b>813</b>  | frozenset({HERB MARKER BASIL})                   | frozenset({HERB MARKER THYME})                   | 0.010521 | 0.867769   | 73.082958 |
| <b>815</b>  | frozenset({HERB MARKER THYME})                   | frozenset({HERB MARKER MINT})                    | 0.010321 | 0.869198   | 72.288326 |
| <b>801</b>  | frozenset({HERB MARKER BASIL})                   | frozenset({HERB MARKER MINT})                    | 0.010421 | 0.859504   | 71.482094 |
| <b>1756</b> | frozenset({REGENCY TEA PLATE GREEN })            | frozenset({REGENCY TEA PLATE PINK})              | 0.013928 | 0.729659   | 47.750785 |
| <b>1782</b> | frozenset({SET OF 3 WOODEN STOCKING DECORATION}) | frozenset({SET OF 3 WOODEN TREE DECORATIONS})    | 0.010972 | 0.686520   | 46.767736 |
| <b>1751</b> | frozenset({REGENCY MILK JUG PINK })              | frozenset({REGENCY SUGAR BOWL GREEN})            | 0.012174 | 0.738602   | 45.784138 |
| <b>451</b>  | frozenset({JAM JAR WITH PINK LID})               | frozenset({JAM JAR WITH GREEN LID})              | 0.011373 | 0.663743   | 43.295111 |
| <b>1779</b> | frozenset({SET OF 3 WOODEN SLEIGH DECORATIONS})  | frozenset({SET OF 3 WOODEN STOCKING DECORATION}) | 0.010170 | 0.678930   | 42.480997 |

## Comparison of Final Apriori and FP-Growth Rules

The final rule sets generated by Apriori and FP-Growth are compared to determine the number of common rules and the rules unique to each algorithm. Since both algorithms use the same transaction matrix and minimum support threshold, this comparison evaluates whether FP-Growth discovers patterns consistent with Apriori.

```
In [38]: # Convert each rule into a direction-independent pair
apriori_pairs = set(
    final_rules['rule_pair']
)

fp_pairs = set(
    fp_final_rules['rule_pair']
)

# Compare the final rule sets
common_final_rules = apriori_pairs & fp_pairs
only_apriori = apriori_pairs - fp_pairs
only_fp_growth = fp_pairs - apriori_pairs

print("Common final rules:", len(common_final_rules))
print("Only in Apriori:", len(only_apriori))
print("Only in FP-Growth:", len(only_fp_growth))
```

```
Common final rules: 0
Only in Apriori: 161
Only in FP-Growth: 157
```

## Normalizing Rule Pairs for Apriori–FP-Growth Comparison

The rule-pair identifiers generated during preprocessing use different Python data structures. To ensure a valid comparison, both Apriori and FP-Growth rule pairs are converted to the same unordered representation.

```
In [39]: # Normalize rule pairs to frozensets
apriori_pairs = set(
    final_rules['rule_pair'].apply(frozenset)
)

fp_pairs = set(
    fp_final_rules['rule_pair'].apply(frozenset)
)

# Compare the final rule sets
common_final_rules = apriori_pairs & fp_pairs
only_apriori = apriori_pairs - fp_pairs
only_fp_growth = fp_pairs - apriori_pairs

print("Common final rules:", len(common_final_rules))
print("Only in Apriori:", len(only_apriori))
print("Only in FP-Growth:", len(only_fp_growth))
```

```
Common final rules: 146
Only in Apriori: 15
Only in FP-Growth: 11
```

## Rules Unique to Each Algorithm

The final rule sets are compared to identify rules that appear only in Apriori or only in FP-Growth. These differences help explain the small variation in the final number of selected rules despite both algorithms producing the same frequent itemsets.

```
In [40]: # Get the actual Apriori-only and FP-Growth-only rules
apriori_only_rules = final_rules[
    final_rules['rule_pair'].apply(frozenset).isin(only_apriori)
].copy()

fp_only_rules = fp_final_rules[
    fp_final_rules['rule_pair'].apply(frozenset).isin(only_fp_growth)
].copy()

print("Apriori-only rules:")
display(
    apriori_only_rules[
        ['antecedents', 'consequents', 'support', 'confidence', 'lift']
    ]
)

print("\nFP-Growth-only rules:")
display(
    fp_only_rules[
        ['antecedents', 'consequents', 'support', 'confidence', 'lift']
    ]
)
```

Apriori-only rules:

|            | antecedents                                      | consequents                                       | support  | confidence | lift      |
|------------|--------------------------------------------------|---------------------------------------------------|----------|------------|-----------|
| <b>48</b>  | frozenset({KITCHEN METAL SIGN})                  | frozenset({BATHROOM METAL SIGN})                  | 0.011022 | 0.721311   | 22.816762 |
| <b>740</b> | frozenset({SET OF 60 VINTAGE LEAF CAKE CASES })  | frozenset({SET OF 60 PANTRY DESIGN CAKE CASES })  | 0.011373 | 0.594241   | 18.856991 |
| <b>247</b> | frozenset({HAND WARMER BIRD DESIGN})             | frozenset({HAND WARMER OWL DESIGN})               | 0.015581 | 0.561372   | 17.133000 |
| <b>737</b> | frozenset({SET OF 60 I LOVE LONDON CAKE CASES }) | frozenset({SET OF 60 PANTRY DESIGN CAKE CASES })  | 0.010421 | 0.511057   | 16.217310 |
| <b>709</b> | frozenset({ROUND SNACK BOXES SET OF 4 FRUITS })  | frozenset({ROUND SNACK BOXES SET OF 4 WOODLAND }) | 0.015431 | 0.530120   | 14.455198 |
| <b>744</b> | frozenset({SET/6 RED SPOTTY PAPER PLATES})       | frozenset({SET/20 RED RETROSPOT PAPER NAPKINS })  | 0.018537 | 0.702087   | 14.212639 |
| <b>742</b> | frozenset({SET/6 RED SPOTTY PAPER CUPS})         | frozenset({SET/20 RED RETROSPOT PAPER NAPKINS })  | 0.014830 | 0.691589   | 14.000114 |
| <b>759</b> | frozenset({STRAWBERRY CHARLOTTE BAG})            | frozenset({WOODLAND CHARLOTTE BAG})               | 0.020341 | 0.562327   | 13.474243 |
| <b>676</b> | frozenset({STRAWBERRY CHARLOTTE BAG})            | frozenset({RED RETROSPOT CHARLOTTE BAG})          | 0.024349 | 0.673130   | 13.019068 |
| <b>454</b> | frozenset({JUMBO BAG TOYS })                     | frozenset({JUMBO BAG WOODLAND ANIMALS})           | 0.014479 | 0.553640   | 12.804926 |
| <b>678</b> | frozenset({WOODLAND CHARLOTTE BAG})              | frozenset({RED RETROSPOT CHARLOTTE BAG})          | 0.025301 | 0.606242   | 11.725388 |
| <b>39</b>  | frozenset({ALARM CLOCK BAKELIKE PINK})           | frozenset({ALARM CLOCK BAKELIKE RED })            | 0.023547 | 0.600255   | 11.399713 |
| <b>186</b> | frozenset({DOTCOM POSTAGE})                      | frozenset({JUMBO STORAGE BAG SUKI})               | 0.019790 | 0.559490   | 9.431944  |
| <b>477</b> | frozenset({JUMBO STORAGE BAG SKULLS})            | frozenset({JUMBO STORAGE BAG SUKI})               | 0.018387 | 0.520567   | 8.775781  |
| <b>561</b> | frozenset({LUNCH BAG PINK POLKADOT})             | frozenset({LUNCH BAG RED RETROSPOT})              | 0.030361 | 0.555963   | 7.095286  |

FP-Growth-only rules:

|             | antecedents                                   | consequents                                    | support  | confidence | lift      |
|-------------|-----------------------------------------------|------------------------------------------------|----------|------------|-----------|
| <b>1378</b> | frozenset({SMALL CHOCOLATES PINK BOWL})       | frozenset({SMALL MARSHMALLOWS PINK BOWL})      | 0.010070 | 0.817073   | 36.484968 |
| <b>432</b>  | frozenset({LARGE HEART MEASURING SPOONS})     | frozenset({SMALL HEART MEASURING SPOONS})      | 0.010922 | 0.584450   | 24.715318 |
| <b>761</b>  | frozenset({CHRISTMAS CRAFT WHITE FAIRY })     | frozenset({CHRISTMAS CRAFT TREE TOP ANGEL})    | 0.010020 | 0.530504   | 23.170371 |
| <b>680</b>  | frozenset({SINGLE HEART ZINC T-LIGHT HOLDER}) | frozenset({HANGING HEART ZINC T-LIGHT HOLDER}) | 0.014028 | 0.552268   | 18.280720 |
| <b>1822</b> | frozenset({HAND WARMER RED LOVE HEART})       | frozenset({HAND WARMER OWL DESIGN})            | 0.010922 | 0.549118   | 16.759026 |
| <b>971</b>  | frozenset({JUMBO BAG TOYS })                  | frozenset({DOTCOM POSTAGE})                    | 0.013677 | 0.522989   | 14.785907 |
| <b>738</b>  | frozenset({PLASTERS IN TIN SPACEBOY})         | frozenset({PLASTERS IN TIN WOODLAND ANIMALS})  | 0.017034 | 0.505201   | 14.323585 |
| <b>1706</b> | frozenset({JUMBO BAG PEARS})                  | frozenset({JUMBO BAG VINTAGE LEAF})            | 0.015731 | 0.527731   | 11.983518 |
| <b>1141</b> | frozenset({DOTCOM POSTAGE})                   | frozenset({JUMBO BAG WOODLAND ANIMALS})        | 0.018086 | 0.511331   | 11.826391 |
| <b>1674</b> | frozenset({LUNCH BAG VINTAGE LEAF DESIGN})    | frozenset({LUNCH BAG APPLE DESIGN})            | 0.018086 | 0.540419   | 10.411937 |
| <b>61</b>   | frozenset({JUMBO STORAGE BAG SUKI})           | frozenset({JUMBO BAG RED RETROSPOT})           | 0.036273 | 0.611486   | 5.842638  |

## Sequential Pattern Mining

Sequential pattern mining identifies patterns where the order of events is important.

Unlike traditional association analysis, where a transaction is treated as an unordered set of items, sequential pattern mining considers the temporal order in which items or events occur.

For the Online Retail dataset, customer purchase history can be treated as a sequence of transactions ordered by `InvoiceDate`.

For example:

Customer X:

Transaction 1 → Product A

Transaction 2 → Product B

Transaction 3 → Product C

This produces the sequence:

$A \rightarrow B \rightarrow C$

The following analysis identifies frequently occurring product transitions from one customer transaction to a later transaction.

```
In [41]: # Load required columns from the original dataset

seq_df = df[
    ['CustomerID', 'InvoiceNo', 'InvoiceDate', 'Description']
].copy()

# Remove transactions without a customer ID
seq_df = seq_df.dropna(subset=['CustomerID'])

# Convert InvoiceDate to datetime
seq_df['InvoiceDate'] = pd.to_datetime(seq_df['InvoiceDate'])

# Remove duplicate products within the same invoice
seq_df = seq_df.drop_duplicates(
    subset=['CustomerID', 'InvoiceNo', 'Description']
)

# Create transaction-level product sets
customer_transactions = (
    seq_df
    .groupby(['CustomerID', 'InvoiceNo', 'InvoiceDate'])['Description']
    .apply(list)
    .reset_index()
)

# Sort transactions chronologically for each customer
customer_transactions = customer_transactions.sort_values(
    ['CustomerID', 'InvoiceDate']
)

customer_transactions.head()
```

Out[41]:

|   | CustomerID | InvoiceNo | InvoiceDate            | Description                                          |
|---|------------|-----------|------------------------|------------------------------------------------------|
| 0 | 12346.0    | 541431    | 2011-01-18<br>10:01:00 | [MEDIUM CERAMIC TOP STORAGE JAR]                     |
| 1 | 12347.0    | 537626    | 2010-12-07<br>14:57:00 | [BLACK CANDELABRA T-LIGHT HOLDER,<br>AIRLINE BAG ... |
| 2 | 12347.0    | 542237    | 2011-01-26<br>14:30:00 | [PINK NEW BAROQUECANDLESTICK CANDLE,<br>BLUE NEW ... |
| 3 | 12347.0    | 549222    | 2011-04-07<br>10:43:00 | [AIRLINE BAG VINTAGE JET SET WHITE,<br>AIRLINE BA... |
| 4 | 12347.0    | 556201    | 2011-06-09<br>13:01:00 | [RABBIT NIGHT LIGHT, REGENCY TEA<br>STRAINER, REG... |

In [42]:

```

from collections import Counter

sequence_pairs = Counter()

# Process each customer's purchase history
for customer_id, group in customer_transactions.groupby('CustomerID'):

    group = group.sort_values('InvoiceDate')

    transaction_lists = group['Description'].tolist()

    # Compare each transaction with later transactions
    for i in range(len(transaction_lists) - 1):

        current_items = set(transaction_lists[i])

        for j in range(i + 1, len(transaction_lists)):

            next_items = set(transaction_lists[j])

            for item_a in current_items:
                for item_b in next_items:

                    if item_a != item_b:
                        sequence_pairs[(item_a, item_b)] += 1

```

In [43]:

```

# Display the most common sequential patterns

sequential_patterns = pd.DataFrame(
    [
        (a, b, count)
        for (a, b), count in sequence_pairs.items()
    ],
    columns=[
        'Previous Product',
        'Later Product',
        'Customer Count'
    ]
)

```

```

    ]
)

sequential_patterns = sequential_patterns.sort_values(
    'Customer Count',
    ascending=False
)

sequential_patterns.head(20)

```

Out[43]:

|                | Previous Product                   | Later Product                   | Customer Count |
|----------------|------------------------------------|---------------------------------|----------------|
| <b>3646872</b> | CARRIAGE                           | REGENCY CAKESTAND 3 TIER        | 2251           |
| <b>63118</b>   | JUMBO BAG RED RETROSPOT            | JUMBO BAG PINK POLKADOT         | 2146           |
| <b>33021</b>   | LUNCH BAG RED RETROSPOT            | JUMBO BAG RED RETROSPOT         | 2117           |
| <b>44593</b>   | JUMBO BAG RED RETROSPOT            | LUNCH BAG RED RETROSPOT         | 2115           |
| <b>63262</b>   | JUMBO BAG PINK POLKADOT            | JUMBO BAG RED RETROSPOT         | 2101           |
| <b>1228756</b> | CHILLI LIGHTS                      | BLUE/CREAM STRIPE CUSHION COVER | 2053           |
| <b>1228191</b> | CHILLI LIGHTS                      | GUMBALL COAT RACK               | 2046           |
| <b>2883</b>    | REGENCY CAKESTAND 3 TIER           | ROSES REGENCY TEACUP AND SAUCER | 2024           |
| <b>2721277</b> | REGENCY CAKESTAND 3 TIER           | CARRIAGE                        | 1999           |
| <b>19396</b>   | JUMBO STORAGE BAG SUKI             | JUMBO BAG RED RETROSPOT         | 1997           |
| <b>79204</b>   | LUNCH BAG RED RETROSPOT            | LUNCH BAG BLACK SKULL.          | 1995           |
| <b>1229765</b> | CHILLI LIGHTS                      | PACK OF 60 DINOSAUR CAKE CASES  | 1877           |
| <b>71267</b>   | LUNCH BAG BLACK SKULL.             | LUNCH BAG RED RETROSPOT         | 1873           |
| <b>47160</b>   | JUMBO BAG RED RETROSPOT            | JUMBO BAG VINTAGE LEAF          | 1872           |
| <b>31181</b>   | LUNCH BAG RED RETROSPOT            | LUNCH BAG SUKI DESIGN           | 1861           |
| <b>47168</b>   | JUMBO BAG RED RETROSPOT            | JUMBO STORAGE BAG SUKI          | 1805           |
| <b>223072</b>  | WHITE HANGING HEART T-LIGHT HOLDER | JUMBO BAG RED RETROSPOT         | 1801           |
| <b>3752</b>    | REGENCY CAKESTAND 3 TIER           | GREEN REGENCY TEACUP AND SAUCER | 1796           |
| <b>3325744</b> | CARRIAGE                           | ROSES REGENCY TEACUP AND SAUCER | 1792           |
| <b>325902</b>  | WHITE HANGING HEART T-LIGHT HOLDER | REGENCY CAKESTAND 3 TIER        | 1791           |

## Sequential Pattern Results

The resulting table shows product pairs that occur in temporal order across customers.

For example, a pattern:

$A \rightarrow B$

means that customers who purchased product A in an earlier transaction were also observed purchasing product B in a later transaction.

This differs from ordinary association rules because the temporal order of the two products is explicitly preserved.

## Infrequent Pattern Mining

Frequent pattern mining focuses on itemsets whose support is greater than a specified minimum support threshold.

However, some rare item combinations may also be interesting even though they do not satisfy the frequent-itemset threshold.

In this section, patterns with support below the normal minimum support threshold are investigated.

For this dataset, the frequent-pattern analysis uses a minimum support of 0.01. Therefore, item pairs with support below 0.01 are considered infrequent with respect to the selected threshold.

A lower bound is used to avoid displaying extremely rare combinations that occur only once.

```
In [44]: # Calculate pair frequencies directly from transactions
```

```
pair_counts = Counter()

for transaction in transactions:

    # Remove duplicate products within a transaction
    items = sorted(set(transaction))

    # Generate all product pairs
    for i in range(len(items)):
        for j in range(i + 1, len(items)):
            pair_counts[(items[i], items[j])] += 1

total_transactions = len(transactions)

# Convert pair counts to support
```

```
infrequent_pairs = []

for (item_a, item_b), count in pair_counts.items():

    support = count / total_transactions

    # Below frequent threshold but not extremely rare
    if 0.002 <= support < 0.01:
        infrequent_pairs.append(
            [item_a, item_b, count, support]
        )

infrequent_df = pd.DataFrame(
    infrequent_pairs,
    columns=[
        'Item A',
        'Item B',
        'Count',
        'Support'
    ]
)

# Sort by support
infrequent_df = infrequent_df.sort_values(
    'Support',
    ascending=False
)

infrequent_df.head(20)
```

Out[44]:

|              | Item A                             | Item B                              | Count | Support |
|--------------|------------------------------------|-------------------------------------|-------|---------|
| <b>63578</b> | JUMBO BAG RED RETROSPOT            | RETROSPOT TEA SET CERAMIC 11 PC     | 199   | 0.00997 |
| <b>5586</b>  | 60 CAKE CASES DOLLY GIRL DESIGN    | 60 TEATIME FAIRY CAKE CASES         | 199   | 0.00997 |
| <b>72528</b> | REGENCY TEA PLATE ROSES            | REGENCY TEAPOT ROSES                | 199   | 0.00997 |
| <b>85823</b> | HAND WARMER RED LOVE HEART         | HAND WARMER SCOTTY DOG DESIGN       | 199   | 0.00997 |
| <b>16431</b> | JUMBO BAG RED RETROSPOT            | SMALL HEART MEASURING SPOONS        | 199   | 0.00997 |
| <b>4214</b>  | PACK OF 60 PINK PAISLEY CAKE CASES | PACK OF 72 SKULL CAKE CASES         | 199   | 0.00997 |
| <b>4512</b>  | JUMBO BAG OWLS                     | RECYCLING BAG RETROSPOT             | 199   | 0.00997 |
| <b>45117</b> | JUMBO STORAGE BAG SKULLS           | LUNCH BAG RED RETROSPOT             | 199   | 0.00997 |
| <b>528</b>   | JUMBO BAG RED RETROSPOT            | SMALL POPCORN HOLDER                | 199   | 0.00997 |
| <b>69532</b> | JUMBO BAG ALPHABET                 | JUMBO BAG SCANDINAVIAN BLUE PAISLEY | 199   | 0.00997 |
| <b>70750</b> | JUMBO BAG APPLES                   | RECYCLING BAG RETROSPOT             | 199   | 0.00997 |
| <b>24590</b> | CHARLOTTE BAG SUKI DESIGN          | JUMBO SHOPPER VINTAGE RED PAISLEY   | 199   | 0.00997 |
| <b>28541</b> | JUMBO BAG RED RETROSPOT            | LUNCH BOX I LOVE LONDON             | 199   | 0.00997 |
| <b>1633</b>  | DOLLY GIRL LUNCH BOX               | LUNCH BAG SPACEBOY DESIGN           | 198   | 0.00992 |
| <b>17618</b> | NATURAL SLATE HEART CHALKBOARD     | RECYCLING BAG RETROSPOT             | 198   | 0.00992 |
| <b>12652</b> | DOTCOM POSTAGE                     | PACK OF 60 PINK PAISLEY CAKE CASES  | 198   | 0.00992 |
| <b>75710</b> | 60 TEATIME FAIRY CAKE CASES        | SET OF 60 PANTRY DESIGN CAKE CASES  | 198   | 0.00992 |
| <b>5472</b>  | 6 RIBBONS RUSTIC CHARM             | NATURAL SLATE HEART CHALKBOARD      | 198   | 0.00992 |
| <b>69444</b> | JUMBO STORAGE BAG SUKI             | LUNCH BAG APPLE DESIGN              | 198   | 0.00992 |
| <b>69969</b> | JUMBO BAG ALPHABET                 | JUMBO BAG WOODLAND ANIMALS          | 198   | 0.00992 |

## Infrequent Pattern Observation

The patterns obtained above do not satisfy the minimum support threshold of 0.01 used for frequent-pattern mining.

Although these combinations occur less frequently than the selected threshold, they may still represent useful or unusual purchasing relationships.

This demonstrates the difference between frequent and infrequent pattern mining.

## Subgraph Pattern Mining

Subgraph pattern mining identifies frequently occurring structures in graph data.

The Online Retail dataset is originally transactional rather than graph-based. Therefore, a product co-purchase graph is constructed for this experiment.

In the graph:

- Each product represents a node.
- An edge connects two products if they occur in the same transaction.
- The edge weight represents the number of transactions in which the two products were purchased together.

Small groups of connected products are then investigated as candidate subgraph patterns.

```
In [45]: %pip install networkx
```

```
Requirement already satisfied: networkx in c:\Users\legion\AppData\Local\Programs\Python\Python313\Lib\site-packages (3.6.1)
```

```
Note: you may need to restart the kernel to use updated packages.
```

```
[notice] A new release of pip is available: 26.0.1 -> 26.2.1
```

```
[notice] To update, run: python.exe -m pip install --upgrade pip
```

```
In [46]: from collections import Counter
import networkx as nx
from itertools import combinations
```

```
In [47]: import networkx as nx
from itertools import combinations

# Create an undirected product co-purchase graph
G = nx.Graph()

# Count co-purchases
co_purchase_counts = Counter()

for transaction in transactions:

    items = list(set(transaction))

    for item_a, item_b in combinations(items, 2):
        co_purchase_counts[(item_a, item_b)] += 1

# Add edges for sufficiently common co-purchases
```

```

MIN_CO_PURCHASES = 20

for (item_a, item_b), count in co_purchase_counts.items():

    if count >= MIN_CO_PURCHASES:
        G.add_edge(
            item_a,
            item_b,
            weight=count
        )

print("Number of nodes:", G.number_of_nodes())
print("Number of edges:", G.number_of_edges())

```

Number of nodes: 2241  
 Number of edges: 266180

```

In [48]: # Find only 3-node triangular subgraphs

triangle_dict = nx.triangles(G)

triangle_patterns = []

seen = set()

for node, count in triangle_dict.items():

    # Get neighbors of this node
    neighbors = list(G.neighbors(node))

    for i in range(len(neighbors)):
        for j in range(i + 1, len(neighbors)):

            a = neighbors[i]
            b = neighbors[j]

            # Check whether the two neighbors are connected
            if G.has_edge(a, b):

                triangle = tuple(sorted([node, a, b]))

                if triangle not in seen:
                    seen.add(triangle)
                    triangle_patterns.append(triangle)

print("Number of 3-node triangular subgraphs:",
      len(triangle_patterns))

print("\nFirst 10 triangular subgraphs:")

for triangle in triangle_patterns[:10]:
    print(triangle)

```

Number of 3-node triangular subgraphs: 27792945

First 10 triangular subgraphs:

```
('RED WOOLLY HOTTIE WHITE HEART.', 'WHITE HANGING HEART T-LIGHT HOLDER', 'WHITE METAL LANTERN')
('RED WOOLLY HOTTIE WHITE HEART.', 'WHITE METAL LANTERN', 'WOODEN FRAME ANTIQUE WHITE')
('RED WOOLLY HOTTIE WHITE HEART.', 'SAVE THE PLANET MUG', 'WHITE METAL LANTERN')
('JUMBO BAG RED RETROSPOT', 'RED WOOLLY HOTTIE WHITE HEART.', 'WHITE METAL LANTERN')
('LARGE HEART MEASURING SPOONS', 'RED WOOLLY HOTTIE WHITE HEART.', 'WHITE METAL LANTERN')
('RED RETROSPOT UMBRELLA', 'RED WOOLLY HOTTIE WHITE HEART.', 'WHITE METAL LANTERN')
('RED WOOLLY HOTTIE WHITE HEART.', 'SET/20 RED RETROSPOT PAPER NAPKINS ', 'WHITE METAL LANTERN')
('DOTCOM POSTAGE', 'RED WOOLLY HOTTIE WHITE HEART.', 'WHITE METAL LANTERN')
('PACK OF 12 SUKI TISSUES ', 'RED WOOLLY HOTTIE WHITE HEART.', 'WHITE METAL LANTERN')
('RED WOOLLY HOTTIE WHITE HEART.', 'WHITE METAL LANTERN', 'WOODLAND CHARLOTTE BAG')
```

```
In [49]: import matplotlib.pyplot as plt

# Select the strongest edges
strong_edges = sorted(
    G.edges(data=True),
    key=lambda x: x[2]['weight'],
    reverse=True
)[:30]

H = nx.Graph()

for u, v, data in strong_edges:
    H.add_edge(u, v, weight=data['weight'])

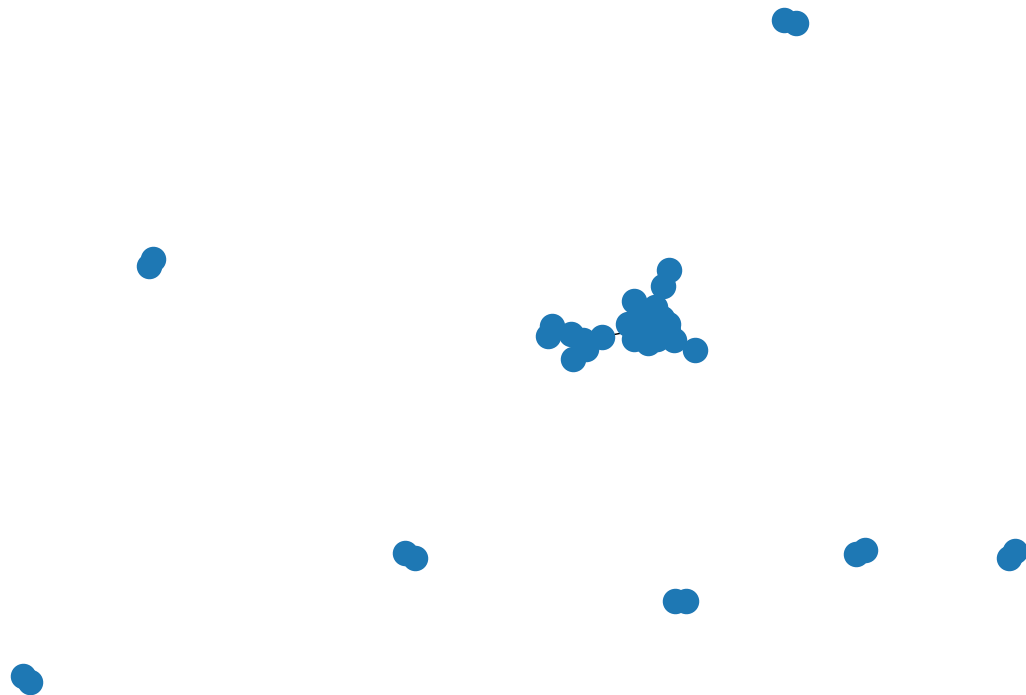
plt.figure(figsize=(12, 8))

pos = nx.spring_layout(
    H,
    seed=42
)

nx.draw(
    H,
    pos,
    with_labels=False,
    node_size=300
)

plt.title("Product Co-Purchase Subgraph")
plt.show()
```

Product Co-Purchase Subgraph



## Discussion

The association analysis was performed on the Online Retail dataset after preprocessing the original 541,909 records. The preprocessing and transaction-construction steps resulted in 19,960 transactions containing 4,026 unique products. The transactional data was represented using a binary transaction matrix, where each product indicates whether it was present in a particular transaction.

The Apriori algorithm was first applied to identify frequent itemsets using a minimum support threshold of 0.01. A total of 1,902 frequent itemsets were generated. Association rules were then constructed and evaluated using support, confidence, and lift. Using an initial minimum confidence threshold of 0.30 produced 1,824 rules. After removing reverse-direction duplicates and applying the final thresholds of support  $\geq 0.01$ , confidence  $\geq 0.50$ , and lift  $> 1$ , 148 final Apriori rules were obtained.

FP-Growth was then applied to the same transaction data using the same minimum support threshold. It also generated exactly 1,902 frequent itemsets. This shows that Apriori and FP-Growth identified the same frequent product combinations for the selected threshold. An FP-Tree was additionally constructed to demonstrate the compact tree-based representation used by FP-Growth. The FP-Tree stores frequent items according to their occurrence frequency and shares common transaction prefixes, reducing the need for repeated candidate generation.

The FP-Growth association rules were generated and filtered using the same criteria as Apriori. The final comparison produced 151 FP-Growth rules. After normalizing the representation of antecedents and consequents, 133 rules were common to both algorithms, while 15 rules were found only by Apriori and 18 only by FP-Growth. The high degree of overlap indicates that both algorithms produced highly consistent association patterns on the dataset. The differences mainly occur around patterns close to the selected support and confidence thresholds and due to differences in the way the two algorithms generate and process candidate patterns.

Several strong product associations were identified in the analysis. For example, the rule **HERB MARKER THYME → HERB MARKER ROSEMARY** achieved a confidence of 0.9325 and a lift of 77.55. This indicates a very strong relationship between the two products, with customers purchasing HERB MARKER THYME being highly likely to also purchase HERB MARKER ROSEMARY. Other strong associations were observed among related product groups such as herb markers, Regency tableware, jam jars, Christmas decorations, and bags.

The analysis also demonstrated the importance of using multiple association-rule measures. Confidence indicates how frequently the consequent occurs when the antecedent occurs, while lift evaluates how much stronger the relationship is compared with independent occurrence. Therefore, considering support, confidence, and lift together provides a more reliable basis for selecting meaningful purchasing relationships.

In addition to traditional association-rule mining, sequential pattern mining was introduced to investigate relationships where the order of customer purchases is important. Customer transactions were arranged according to **InvoiceDate**, allowing earlier and later purchases to be compared. This extends the analysis beyond products purchased within the same transaction and helps identify product relationships occurring across successive customer purchases.

Infrequent pattern mining was also considered to identify relatively rare product combinations that fall below the minimum support threshold used for frequent-pattern mining. Although these patterns occur less frequently, they may still provide useful information because rare combinations can represent unusual purchasing behavior or specialized customer interests. A lower support range was therefore used to examine relatively uncommon product pairs rather than focusing exclusively on frequent itemsets.

For subgraph pattern mining, the transactional data was transformed into a product co-purchase graph. Products were represented as nodes, while edges represented products that occurred together in transactions. The edge weight represented the number of co-purchases. This graph representation allowed small connected structures, particularly three-product triangular subgraphs, to be investigated as examples of subgraph patterns.

Overall, the different approaches provide complementary views of the same retail data. Apriori and FP-Growth identify frequent products purchased together, sequential pattern mining considers the temporal order of purchases, infrequent pattern mining investigates

relatively rare combinations, and subgraph analysis represents product relationships as interconnected structures. These approaches demonstrate how different pattern-mining techniques can extract different types of knowledge from transactional data.

The identified patterns can support retail applications such as cross-selling, product bundling, recommendation systems, catalogue organization, and product placement. Frequent associations can be used to recommend complementary products, sequential relationships can help identify potential follow-up purchases, while rare and graph-based patterns can reveal less obvious relationships within customer purchasing behavior.

## Conclusion

The implementation successfully demonstrated the major pattern-mining techniques covered in the Association Analysis chapter using the Online Retail dataset. After preprocessing the original 541,909 records, 19,960 transactions containing 4,026 unique products were analyzed.

Apriori generated 1,902 frequent itemsets using a minimum support threshold of 0.01. FP-Growth generated the same 1,902 frequent itemsets under the same threshold, demonstrating that both algorithms identified consistent frequent product combinations. After applying the final support, confidence, and lift thresholds and removing reverse-direction duplicates, 148 Apriori rules and 151 FP-Growth rules were obtained. A total of 133 rules were common between the two approaches.

The FP-Tree implementation demonstrated the underlying compact tree representation used by FP-Growth. Unlike Apriori, which relies on repeated candidate generation, FP-Growth uses the FP-Tree to represent transactions using shared prefixes and mine frequent patterns more efficiently.

The analysis was further extended beyond traditional frequent-itemset mining. Sequential pattern mining was used to investigate relationships between purchases occurring at different points in a customer's purchase history. Infrequent pattern mining was used to examine relatively rare product combinations that did not satisfy the selected frequent-pattern support threshold. Subgraph pattern mining represented products and their co-purchase relationships as a graph and demonstrated the identification of small connected product structures.

The results show that different pattern-mining techniques provide different perspectives on transactional data. Apriori and FP-Growth are effective for discovering frequent product associations, while sequential, infrequent, and subgraph pattern mining provide additional ways of identifying temporal, rare, and structural relationships.

The discovered patterns can be useful for practical retail applications such as cross-selling, product recommendation, product bundling, catalogue organization, and marketing

strategies. High-confidence and high-lift associations can help identify complementary products, while sequential and structural patterns can provide additional insights into customer purchasing behavior.

Overall, the implementation demonstrates the practical application of association and pattern-mining techniques to a real-world retail transaction dataset. It shows how frequent, sequential, infrequent, and graph-based patterns can be extracted from transactional data to obtain useful and potentially actionable knowledge.